Aspira Edge

2026-27

CONCEPTS | CLINICALS | MCQS

MICROBIOLOGY

- Micro to Maximum Marks.
- Organism-wise quick revision tables.
- Lab diagnosis, toxins & frequently asked points.

DR. KEYUR GANDHI DR JITENDRA

MICROBIOLOGY

GENERAL MICROBIOLOGY

All cocci are gram positive except- VENOM
Veillonella,Neserria (gonococci/meningococci), Moraxella

All bacilli are gram negative except MAC DoNALD
M- Mycobacterial species
A- Anthracis bacillus
C- Clostridium species
D- Diphtheriae corynebacterium
N- Nocardia
A- Actinomyces
L- Listeria
D- Diphtheroids

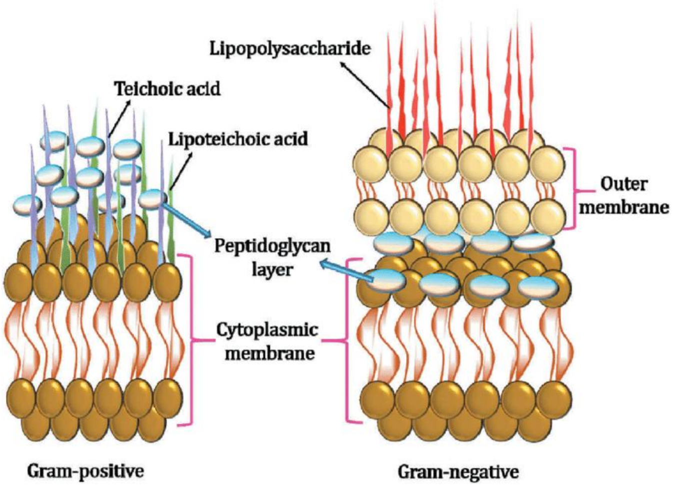
STAINING

1. Gram staining

Sequence- trick- methyl GAS

Methyl violet
↓
Gram's iodine (fixer)
↓
Acetone (decolouriser)
↓
safranin(counter stain)

Gram positive- Purple/violet
Gram negative- Pink



LPS is a feature of gram negative bacteria **FMGE 2022**
Teichoic acid is more in gram positive bacteria **NEET PG 25**
Teichoic acid helps to bind to fibronectin on host epithelial cells → facilitate initial colonisation
All gram -ve release exotoxin , one gram +ve listeria also release exotoxin

2. Acid fast staining (ZN staining) **FMGE 2021,23**
Trick- CHAMP
Carbol fuchsin → Heat → Acid (H_2SO_4) → methylene blue

Mycobacterium Tb - 20% acid fast
M.laprae - 5% acid fast
Nocardia,legionella - 1% acid fast

NEET PG 23

Structure	**Function**
Flagella	Motility
Fimbriae	Help in adhesion
Capsule	Prevent from phagocytosis

BIPOLAR STAINING → safety pin appearance **FMGE 2021**
Yersinia pestis , Donovanosis,Vibrio parahaemolyticus,Haemophilus ducreyi,Burkholderia pseudomallaei

NOTE- Capsule is made of polysaccharide except bacillus anthracis made up of polypeptide.
Quellung reaction-Capsular swelling- Eg- Haemophilus influenzae, Pneumococcus
Mcfadyen reaction- Bacillus Anthrax

MOTILITY

1. Falling leaf like- Giardia
2. Twitching- Trichomonas vaginalis
3. Swarming- Proteus,vibrio parahaemolyticus
4. Tumbling- Listeria (differential motility eg- listeria and yersinia)
5. Darting- V. cholera
6. cork screw - Treponema pallidum

TRANSPORT MEDIA

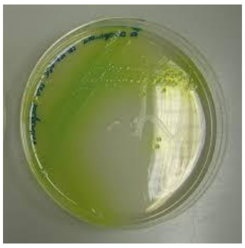
1. Stuart's Amies media- Gonococci and Meningococci
2. Pikes media- S. Pyogenes
3. Cary Blair media- V. cholera

UREASE POSITIVE → tick - PUNCH KISS
Proteus
Ureaplasma
Nocardia
Cryptococcus
H pylori
Klebsiella
Saprophyticus

Mac CONKEY AGAR **FMGE 2021,23**
use for gram negative bacilli
Lactose fermenter
Positive - pink color
Eg-
E.COll
KLEBSIELLA
ENTEROBACTERIA
CITROBACTER

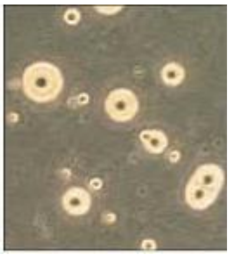
POTASSIUM TELLURITE AGAR - C.DPITHERIAE

CETRIMIDE AGAR - PSEUDOMONAS

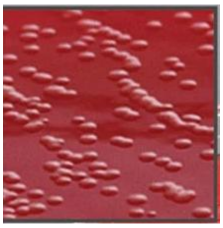


BUFFERED CHARCOAL YEAST EXTRACT - LEGIONELLA

PLEURO-PNEUMONIA LIKE ORGANISM - MYCOPLASMA



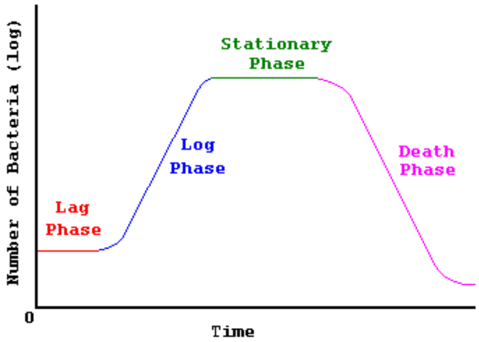
SKIRROW MEDIA - CAMPHYLOBACTER



Bacterial virulence factors

Capsular polysaccharide: Prevents phagocytosis; main virulence factor.
Protein A (S. aureus): Binds Fc of IgG → blocks opsonization & phagocytosis.
IgA protease (SHiN bacteria): Cleaves IgA → helps bacteria stick to mucosa.
S pneumoniae, H influenzae type b, and Neisseria
M protein (Group A Strep): Prevents phagocytosis; cross-reacts with heart tissue → rheumatic fever.

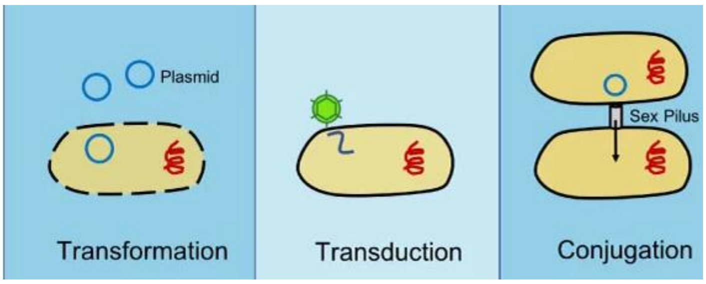
Bacterial growth curve- lag phase, log phase, stationary phase, decline phase



1. Lag phase- increase size, increase enzymes, metabolites.
2. Log phase/ exponential- max activity, uniform staining.
3. Stationary phase- STA- sporulation, toxin production, antibiotic,viable but non culturable cell seen
4. Decline- Death → complete accumulation of toxic metabolites

BACTERIAL GENETICS

FMGE 26



TRANSFORMATION - naked DNA taken up from the environment by bacterial cells
TRANSDUCTION - use of bacteriophage to transfer DNA b/w cells

CONJUGATION - direct transfer of DNA from one bacterial cell to another

Note - antibiotic resistance shown by staph aureus by transduction

STERILISATION AND DISINFECTION

Dry heat- 2 types- Hot air oven, Incineration **FMGE 2022**

1. Hot air oven- 160*c over 2 hours
Used for glasswares, liquid paraffin, syringes, cotton swab,scalpel

2. Incineration- 1200*c
Used for hospital waste(human body part),human anatomical waste, animal carcasses, soiled bedding, dressings, expired cytotoxic drugs.

Control used- Bacillus subtilis, bacillus atrophaeus or c. tetani

Moist heat

<100*c- Pasteurization and Inspissation

1. Pasteurisation-
Holder method - 60°c for 30 min
Flash method - 72°c for 15-20 sec → bring down to 13°c → kills coxiella burnetii spores also

2. Inspissation- 80*C for 20-30 min for 3 consecutive days.

Vaccine water bath- 60*c for 60 min, bacterial vaccines are heat inactivated

@100*c- BAT

1. Boiling: At 100° for 15 min
Does not kill spores
They came up with tyndallization

2. Tyndallization: 100°c x20min x 3days
Kill spores

>100*c- Autoclave → 121°c x15 min x15 psi
Looks like a pressure cooker If you want to kill prions- 134°c x 1.5 hrs
USES- Instruments, Aprons, Sutures, except Catgut Sutures (heat sensitive)
Control- Bacillus Stearothermophilus

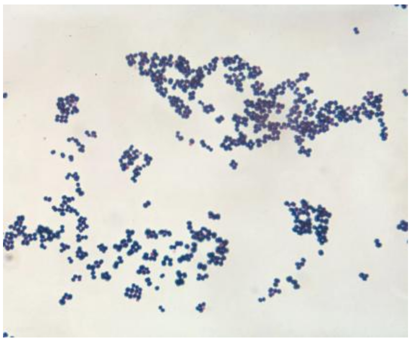
IONIZING RADIATION **FMGE 2021**
Gamma rays/x rays
Breaks upon DNA without temperature rise
Uses- catgut sutures

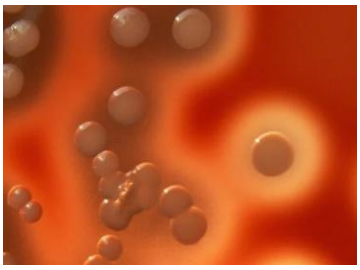
ISOPROPYL ALCOHOL
Used for stethoscope and thermometer x 15 min

GLUTARALDEHYDE
2% glutaraldehyde → CIDEX
Uses → Bronchoscopes,Endoscopes

PLASMA STERILISATION - uses hydrogen peroxide **FMGE 2023**
Uses → Arthroscopes

SYSTEMIC BACTERIOLOGY
STAPHYLOCOCCUS
1. Staphylococcus aureus **FMGE 2022,25**
Catalase +,Coagulase +
Grape like clusters
Best selective agar- Mannitol salt agar
Blood agar -causes ß hemolysis → complete zone of hemolysis





Toxin mediated disease
1. Food poisoning-
-Due to Enterotoxin A

-IP- 6 hrs
-Cause- contaminated meat and milk products

Note: Bacillus cereus - due to chinese food

2. SSSS
Due to exfoliative/enterotoxin, toxin A and B

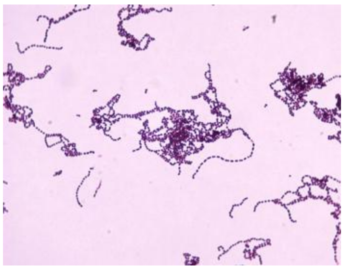
3. Toxic shock syndrome **FMGE 2025**
Due to use of vaginal tampons
Due to enterotoxin F
It is a superantigen

Disease caused by s.aureus- acute osteomyelitis, acute mastitis,pneumatocele,boromycosis
MCC of early prosthetic valve endocarditis or native valve endocarditis.
Most common method of Resistance transfer- Transduction

NEET PG 24

Staph. saprophyticus- most common gram positive coagulase negative bacteria staph causing UTI

STREPTOCOCCUS
GROUP A STEPTOCOCCUS
1. Streptococcus pyogenes **FMGE 2023,24**
Catalase -ve
Selective media- polymyxin B sulfate, neomycin sulfate, and fusidic acid
Transport media- Pike's media



Diseases caused by pyogenes - necrotising fasciitis,erysipelas
Complications
→ Acute Rheumatic
Fever History of sore throat

→ Acute glomerulonephritis
History of pyogenic infection/sore throat

GROUP B STEPTOCOCCUS **FMGE 2023**
Strepto. Agalactiae- mcc of neonatal meningitis

GROUP C STEPTOCOCCUS

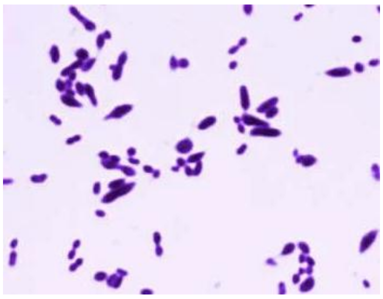
ENTEROCOCCUS
Resist -
6.5% NACL
pH-9.4
40% bile
Cause left sided valve endocarditis

ALPHA HEMOLYTICUS

1. Streptococcus viridans-
MCC of SABE(subacute bacterial endocarditis)
BIO -ve (bile soluble,inulin fermenter,optochin sensitivity)

2. Streptococcus mutans- leads to dental caries

3. Streptococcus pneumoniae
 Diplococci
 BIO +ve
 Caram coin appearance of colonies → Draughtman appearance

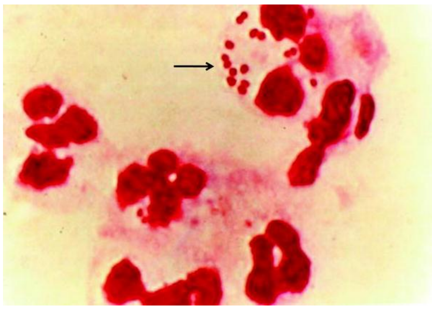


Diseases-
Otitis media
Meningitis → most common cause of meningitis in adults
Pneumonia → most common cause of community acquired pneumonia (x ray finding - lobar consolidation and bronchial breathing)

MENINGOCOCCI

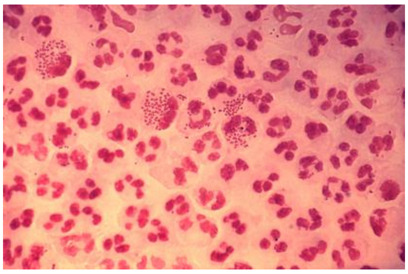
FMGE 2021,25,NEET PG 24

Catalase + and oxidase +
Capsulated
Lens shaped
Ferments glucose and maltose
Culture media - THAYERMARTIN MEDIA
Causes-
meningitis(rash+fever+headache+vomiting+neck rigidity)
Complications-
-Water house friderichsen syndrome



GONOCOCCI **NEET PG 25**

Catalase and oxidase +ve
Kidney shape
Ferment glucose only
Culture media - THAYERMARTIN MEDIA

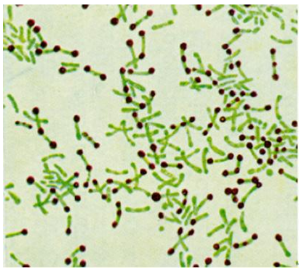


Complications-
WATER CAN PERINEUM
FITZ HUGH CURTIS SYNDROME

CORYNEBACTERIUM DIPHTHERIA **FMGE 2021,24**

-Stain by PLAN (Ponders stains, Loeffler's serum, Albert stain, Neisser stain)
-Chinese letter pattern

DISEASE
Pseudomembrane (greyish whitish membrane)

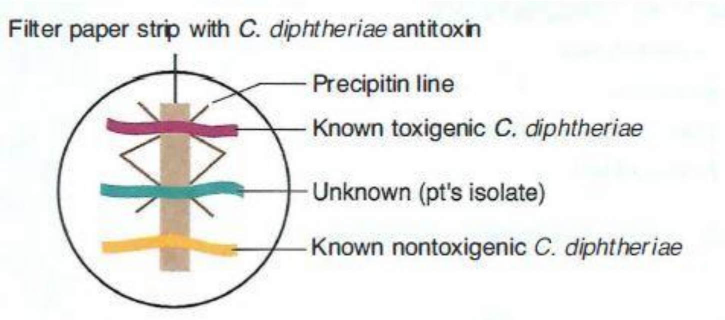


Lab diagnosis-
Loeffler's serum slope → for early diagnosis
Potassium tellurite agar → Definitive Diagnosis Best selective media
→ Takes 48 hrs to grow

Decreasing protein synthesis
D- Diphtheria P-Pseudomonas S- Shiga toxin
Diphtheria and Pseudomonas → Inactivate elongation factor (EF-2) through ADP-ribosylation
Shigella spp → Inactivate 60S ribosome by removing adenine from rRNA

TOXIN DEMONSTRATION

Elek gel precipitation test - to know the toxigenicity of C. diphtheriae



Diphtheria toxin- Bipartite toxin (toxin A+B)
Mechanism- ADP ribosylation of EF-2 + Binding

DIPHTHERIA IS A TOXEMIA NOT A BACTEREMIA

BACILLUS ANTHRACIS **+ FMGE ,2026 2025,24,23,NEET PG 25**

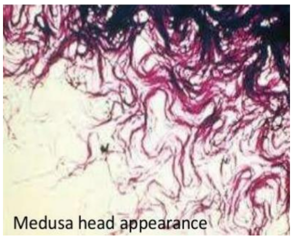
Capsule made up of polypeptide
Anthrax toxin (tripartite toxin)
E- Edema
P- Protective factor
L- Lethal factor

↑ CAMP level
C-Cholera
A-Anthrax
M→ E- E-coli labile toxin
P-Pertussis

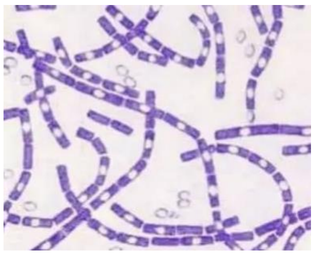
↑ CGMP → E coli stable toxin

Culture media - PLET media (polymyxin lysozyme EDTA thallous acetate)
Shows Mc fadyean reaction

DISEASES
Cutaneous anthrax aka HIDE PORTER DISEASES
black eschar → edema Pulmonary anthrax
Lab diagnosis- PCR for bacillus
Medusa head appearance



Inverted fir tree appearance



String of pearl appearance
Bamboo stick appearance
Trt - Doxycycline

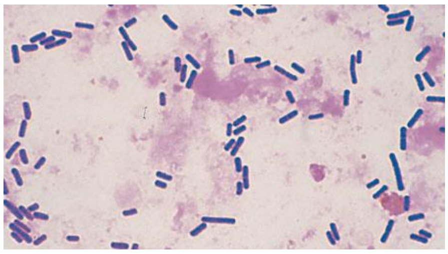
BACILLUS CEREUS

FMGE 2022

Causes food poisoning
IP- 6 hrs
Due to ingestion of chinese food
Culture media- MYPA is cereus (mannitol yolk sac polymyxin phenol red agar)

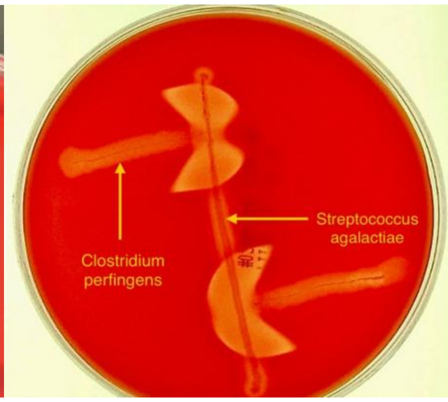
CLOSTRIDIUM PERFRINGENS

FMGE 2025

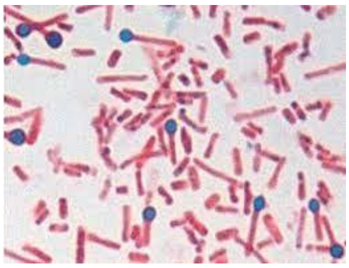


Causes Gas gangrene
IP 1-2 days
C/F- Pain, Crepitus, H/O trauma given
Media- Robertson cooked meat media
Secrete Alpha toxin → Phospholipase (lecithinase) that degrades tissue and cell membranes

Reactions (Senior NTR)
S- Stormy fermentation of litmus milk
N- Nagler reaction
T- Target hemolysis
R- Reverse camp test



CLOSTRIDIUM TETANI



Drumstick appearance
Tetanospasmin- Neurotoxin
Causes-presynaptic inhibition of glycine/GABA (inhibitory neurotransmitter)
NOTE- Strychine poisoning- postsynaptic inhibition of glycine/GABA

CLOSTRIDIUM BOTULINUM

FMGE 2025,2026

Enterotoxin
Due to canned food poisoning
Canned food poisoning
Food act as a passive vehicle born transmission
Symptoms → Dysphagia, Dysarthria, Diplopia, Constipation, ↓ deep tendon reflexes
Botulinum toxin - bind to Ach receptor → flaccid paralysis

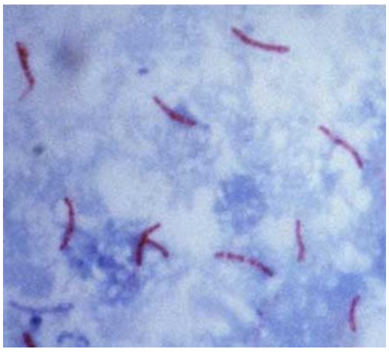
CLOSTRIDIUM DIFFICILE **FMGE 2024,NEET PG 22**

associated with diarrhea
Ulcer- Volcano eruptions
Due to long term use of clindamycin and 3rd generation cephalosporins
Toxin assay best test for detection

MYCOBACTERIUM TUBERCULOSIS **FMGE 2022,24,25**

20% acid fast
Zn staining uses
Culture on LJ media- Rough Tough and Buff colonies

CBNAAT- for Mycobacterium TB + rifampicin resistance
Auramine-rhodamine staining- fluorescent staining
Latent TB- interferon gamma release assay (IGRA)
Vaccine- BCG → derived from danish 1331 strain of M. Bovis



ATYPICAL MYCOBACTERIA

PHOTO CHROMOGENS - grows in light
Marinum mycobacterium
Kansasii mycobacterium

SCOTO CHROMOGENS - grow in darkness
Szulgai mycobacterium
Scrofulaceaum mycobacterium

NON CHROMOGENS - neither grown in darkness and in light
Mycobacterium avium complex
Mycobacterium ulcerans
RAPID GROWERS

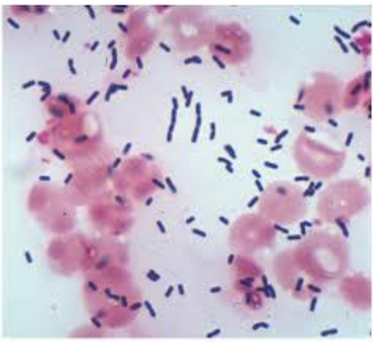
Mycobacterium cheloni
Mycobacterium smegmatis

MYCOBACTERIUM LAPRAE — FMGE 2022

5% acid fast
Does not follow KOCH postulates, does not cultured in pure culture media
Grows in foot pad of Armadillo

LISTERIA MONOCYTOGENES — FMGE 2021

Tumbling motility
Differential motility - at 37°c → non motile
- at 20-25°c → motile



Listeria - Actin polymerisation for intracellular movement
Culture - PALCAM agar
In pregnancy - avoid refrigerated food
↓
Contain listeria → causes premature rupture of membrane

Reactions
ANTON TEST
CAMP TEST
CATALASE TEST

E.COLI — FMGE 2021

Diarrhea causing E.COLI
EPEC- seen in Paediatric age group
EIEC- Invasive- dysentery→resembles shigellosis
ETEC-seen in Travelers
EHEC- causes HUS and Hemorrhagic colitis
EAEC- Persistent diarrhea (>2 weeks)

KLEBSIELLA PNEUMONIA

H/O-Chronic alcoholic, Red currant jelly sputum → typical pneumonia

SALMONELLA TYPHI — FMGE 2022,23,25

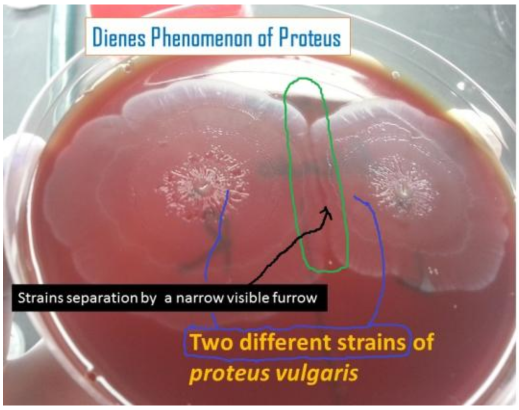
Causes typhoid fever
Longitudinal ulcers
Complications- Pea soup diarrhea

Lab diagnosis- BASU
B-blood culture - 1st week
A-agglutination test - 2nd week (widal test)
S-stool culture -3rd week
U-urine culture - 4th week

Best selective media- Wilson and blair media

PROTEUS **FMGE 2025**

Swarming motility seen
Can be cultured on - phenyl pyruvic acid agar
Fishy smell produced from media
Dienes phenomenon +



Associated with struvite stones
Produces urease
Proteus antigen Ox19,2&k cross react with rickettsial family member → give WEIL-FELIX reactions

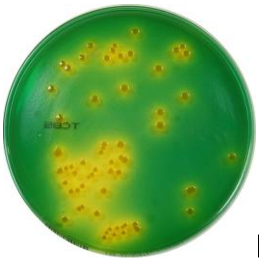
PSEUDOMONAS **FMGE 2025**

Catalase +, oxidase +


Best selective media → cetrimide agar

VIBRIO CHOLERA **FMGE 2023,24 26,NEET PG 22**

Toxin A - ADP ribosylation of GTP + B helps in binding to GM1 ganglioside receptor
Hanging drop motility
Fish in a stream appearance - Stool sample → Rice watery stool
Comma shape,darting motility


best selective media - TCBS agar

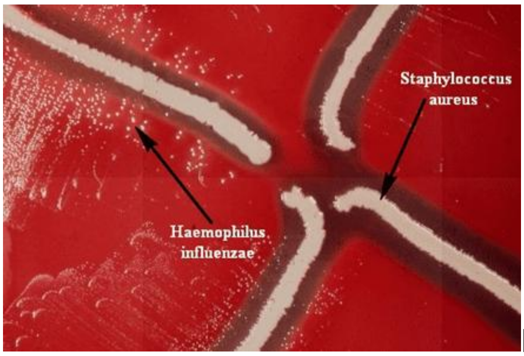
V. PARAHAEMOLYTICUS **FMGE 2023,NEET PG 24**

cause Sea food poisoning
Associated with kanagawa phenomenon on wagatsuma agar

HAEMOPHILUS DUCREYI **FMGE 2025,2026**

Causes chancroid (painful,soft genital ulcer)
School of fish appearance is seen
Culture media → chocolate agar with ISOVITALE X

HAEMOPHILUS INFLUENZAE **FMGE 2025,NEET PG 23**



It shows Satellitism

BACTERIA WHICH DO NOT STAIN ON GRAM STAIN
M- MYCOPLASMA
R- RICKETTSIA
C- CHLAMYDIA
S- SPIROCHETES

Spirochetes contains → endoflagella → in periplasmic space

TREPONEMA PALLIDUM **FMGE 2021,23,NEET PG 21**

Causes syphilis (painless)
IP- 9-90 days Stages
Primary stages - Hard painless chancres

Secondary stages - Condyloma lata, two hand two feet rashes
Tertiary stages -
Gummas seen
Tabes dorsalis
Tree bark aorta
Lab diagnosisMicroscopy- Dark field
IOC for neurosyphilis- VDRL- CSF

Tricks
CAT SAT on MAT
MY BIL LI
Cold agglutination test - Mycoplasma
Standard agglutination test - Brucellosis
Microscopic agglutination test- Leptospira

LEGIONELLA PNEUMONIA

Mode of transmission → aerosol route from centralised AC
Culture media - BCYE
Diseases - PONTAIC FEVER ,leggionare pneumonia

BORRELIA BURGORFERI

Transmitted by ticks
Causes lyme diseases
Characteristic lesion → Annular bulls eye pattern rash

LEPTOSPIRA

FMGE 2025,NEET PG22, 23,24 ,26,26

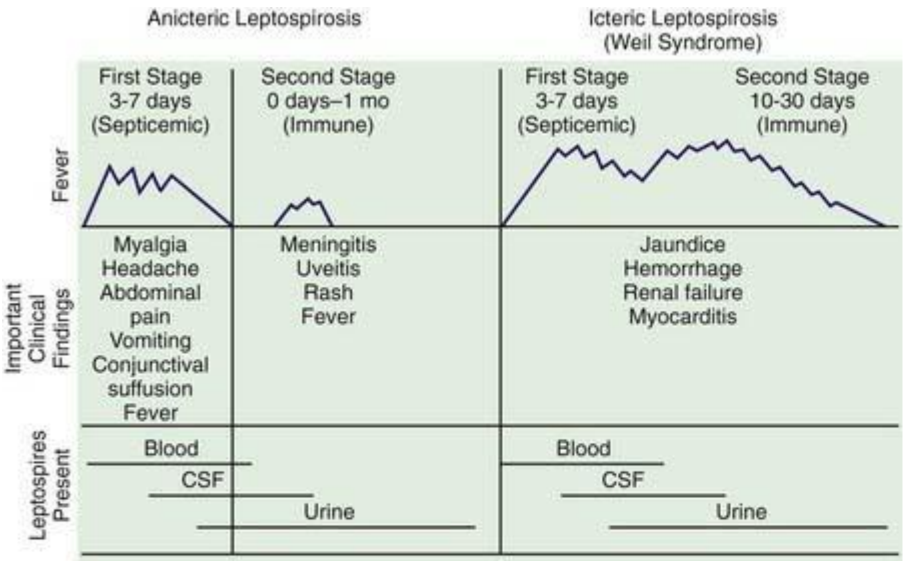
Causes weils diseases

Mode of transmission -
Rat urine Rice Rainy water

Clinical stages

First stage	Second stage
Fever ,myalgia,conjunctival Suffusion, abdominal pain	Meningitis,uveitis,chorioretinitis,rash,fever

Serological test - MAT - microscopic agglutination test


Anicteric Leptospirosis
Icteric Leptospirosis
(Weil Syndrome)
First Stage
3-7 days
(Septicemic)
Second Stage
0 days–1 mo
(Immune)
First Stage
3-7 days
(Septicemic)
Second Stage
10-30 days
(Immune)
Fever
Important Clinical Findings
Myalgia
Headache
Abdominal pain
Vomiting
Conjunctival suffusion
Fever
Meningitis
Uveitis
Rash
Fever
Jaundice
Hemorrhage
Renal failure
Myocarditis
Leptospires Present
Blood
CSF
Urine
Blood
CSF
Urine

RICKETTSIAE

FMGE 2024,NEET PG 22

	CAUSED BY	Arthropodes involve	TRICKS
Epidemic **T**yphus	R. Prowazekii	**L**ouse	**L ET**
ENdemic **T**yphus	R. Typhi	**F**lea	**F EN**
RMSF	R. Rickettsii	**T**ick	**T RIA**
Indian tick typhus	R. Conorii	**T**ick	**T RIA**
R. **P**ox	R. Akari	**M**ite	**PS M**
Scrub typhus-eschar formation	Orientia tsutsugamushi	**M**ite	**PS M**

C/F- fever,rash,lymphadenopathy

Rashes trunk → extremities except RMSF - extremities → trunk

Q FEVER

FMGE 2025

Caused by coxiella burnetii

No rash occurs

Non cultured

Associated with negative endocarditis

EHRLICHIA CHAFFEENSIS

Causes human monocyte ehrlichiosis

EHRLICHIA EWINGL

Causes human granulocytic ehrlichiosis

CAMPHYLOBACTER

NEET PG 25

Darting motility

Gull wing shape

Causes contamination of poultry products

Causes diarrhea → f/b → Guillain Barrie Syndrome

Cultured on - Skirrow media

MYCOLOGY

Classification based on morphology

1. Yeast / True yeast → Cryptococcus neoformans
2. Yeast like → Candida albicans
3. Moulds → Rhizophus, Aspergillus
4. Dimorphic fungi → yeast at 37°c
   mold at 22°c

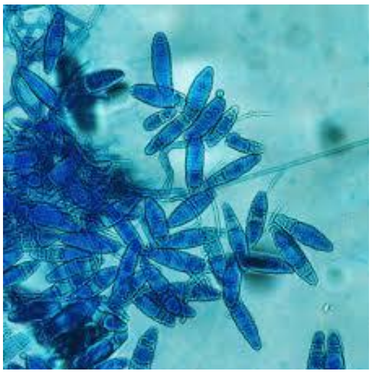
HBSP2C

Histoplasmosis | Blastomycosis | Sporotrichosis
Paracoccididomycosis | Penicilliosis | Coccididomycosis

TINEA VERSICOLOR **FMGE 2021**

Caused by Malassezia Furfur- seen both hypo and hyper pigmented lesions
Diagnosis-Spaghetti and meatball appearance seen

DERMATOPHYTES **FMGE 2023,24**

TRICHOPHYTON	MICROSPORUM	EPIDERMOPHYTON
Skin, Hairs, Nails	Skin, Hairs	Skin, Nails
Cylindrical shape	Spindle shape	Club/boat shape
Microconidia +++	Microconidia ++	Microconidia absent
		

ECTOTHRIX	ENDOTHRIX
Sheats of spores around the shaft	Arthroconidia within the hair shaft
Microsporum canis	Trichophyton tonsurans Trichophyton violacaeum

NEET PG 21

KERION	FAVUS
Boggy swelling over scalp	Crust over scalp
Trichophyton mentagrophytes	T. schoenleinii

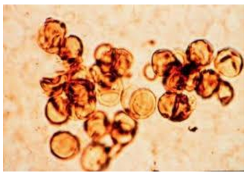
	

CHROMOBLASTOMYCOSIS **FMGE 2022**

Due to barefoot walking→ thorn pick/wood prick

Leads to Verrucous lesion

Biopsy shows → Copper penny bodies, sclerotic medlar bodies



SPOROTRICHOSIS **FMGE 2023,25**

Spread by thorn pick of rose

Lymphocutaneous spread is seen

It is Dimorphic fungi

-mold - flower like sporulation

-yeast - cigar shape yeast cells

Asteroid bodies is seen

HISTOPLASMOSIS **NEET PG 25**

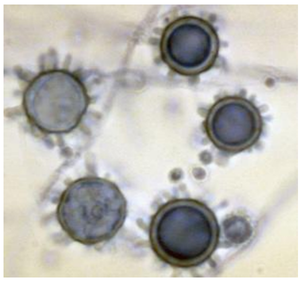
Macrophage filled with Histoplasma → affect reticuloendothelial system

Associated with bird or bat droppings (eg, caves)

Diagnosis via urine/ serum antigen

Mold - tuberculate macroconidia

Yeast - narrow based budding yeast cell



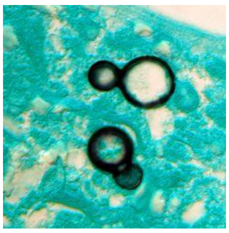
BLASTOMYCOSIS **NEET PG 24**

Broad-based budding of Blastomyces

Disseminates to bone/ skin

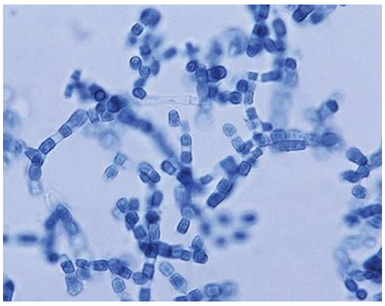
Yeast form - broad based budding

Figure of 8 appearance



COCCIDIOIDOMYCOSIS

-Spherule filled with endospores of Coccidioides
-Arthralgias (desert rheumatism)
-Can cause meningitis
-Erythema nodosum (desert bumps)
-Barrel shape



PARACOCCIDIOIDOMYCOSIS

Common in coffee growing belt
Budding yeast of Paracoccidioides with "captain's wheel" formation

mariner's wheel/mickey mouse appearance

CANDIDA ALBICANS

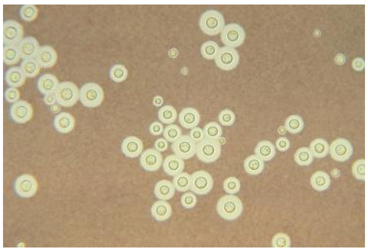
FMGE 2025,NEET PG 24,22,26

Shows Reynold Braude phenomenon → germ tube formation
Culture on SDA → creamy,pasty colonies
Chrom agar → colorful agar
ß glucan test for invasive candidiasis
Vaginal candidiasis- show shish kebab effect on pap smear

CRYPTOCOCCUS NEOFORMANS

FMGE 2023,24,25,26

MCC of meningitis in HIV +ve patient
Lab diagnosis - Indian ink staining → negative staining
On niger seed agar → blackish brownish colonies
Test - Lateral flow assay >> Latex card agglutination test



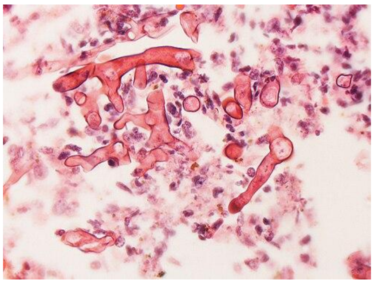
Trt- LAMB+flucytosine

MUCORMYCOSIS

FMGE 2021

Right angle/obtuse angle branching

Aseptate

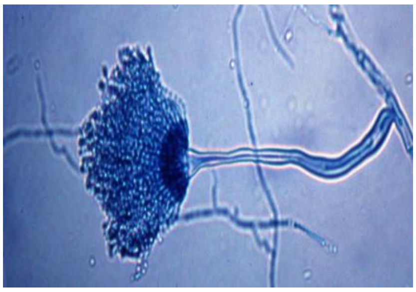


ASPERGILLUS

FMGE 2023,24,NEET PG 25

Acute angle branching

Septate



PNEUMOCYSTIS CARINII

FMGE 2024,25,NEET PG 25

Occur in HIV +ve patient with CD4 count < 200

Ping pong crushed cell seen

Plasma cell present

Lab diagnosis → Bronchoalveolar lavage → due to dry cough

PROTOZOOLOGY

ENTAMOEBA HISTOLYTICA

FMGE 2024,25

Amebiasis—bloody diarrhea (dysentery), liver abscess (“anchovy paste” exudate)

Erythrophagocytosis seen → characteristic feature

Causing Flask shaped ulcers

GIARDIA LAMBLIA FMGE 2021,23

Mostly in duodenum → Malabsorption syndrome
Tennis racket appearance or Tear drop appearance
Axostyles are present
Shows falling leaf like motility
String test positive
Giardiasis—bloating, flatulence, foul-smelling, nonbloody, fatty diarrhea

TRICHOMONAS VAGINALIS FMGE 2022,NEET PG 23

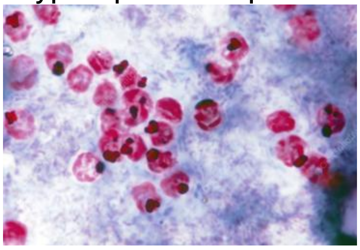
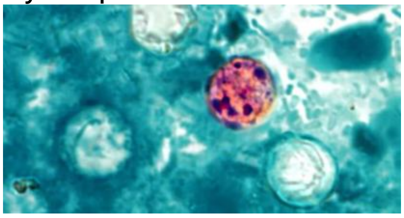
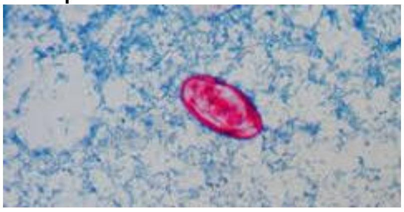
Only trophozoite stage
Twitching motility
Culture on- lash and diamond media

T. CRUZI

Transmitted by- Reduvid bug
Causes Chagas disease
Romana sign seen
Megaesophagus, Megacolon, Myocarditis
Treatment- Benznidazole

COCCIDIAN PARASITE **FMGE 2021**

Diarrhea in HIV +ve patient

Parasite	Oocyte size	Treatment
Cryptosporidium parvum 	4-8 μ	Nitazoxanide
Cyclospora 	8-10 μ	Cotrimoxazole
Isospora 	23-36 μ	Cotrimoxazole

PLASMODIUM SPECIES **FMGE 2021,24,NEET PG 23**

PLASMODIUM VIVAX

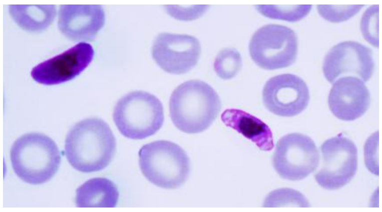
1 Ring 1 Dot trophozoites

Shuffner dots seen

Gametocyte- oval

Hypnozoites- responsible for relapse

DOC- Primaquine

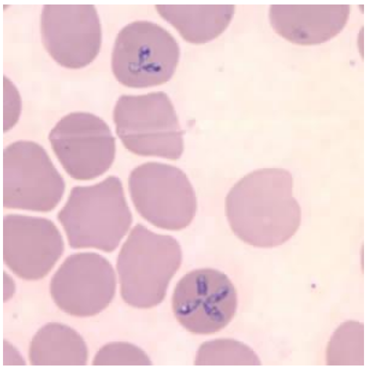


PLASMODIUM FALCIPARUM

Maurer dots seen
Gametocyte- Banana shape
Card test - HRP-2 kit

BABESIA MICROTI

Intraerythrocytic protozoa
Transmitted by tick
Maltese cross arrangement in RBC



FMGE 2023

Trematodes	Definitive host	First intermediate host	2nd intermediate host
Schistosoma	man	snail	–
Paragonimus westermani	man	snail	Crab (para=cara)
Clonorchis sinensis	man	snail	Fish (clonorfish)
Opisthorchis	man	snail	Fish
Fasciola hepatica	man	snail	Fresh aquatic vegetation

NEET PG 23

Trematodes	Infective form	transmission	Adult worms	Eggs	DOC
schistosoma	Cercaria larva	skin	Sexes separate	Spines	Praziquantel
Paragonimus westermani	metacercaria	ingestion	hermaphrodite	Operculated egg	Praziquantel
Clonorchis	metacercari	ingestio	hermaphrodit	Operculated	Praziquantel

(cholangiocarcin oma)	a	n	e	egg	
opisthorchis	metacercari a	ingestio n	hermaphrodit e	Operculated egg	Praziquantel
Fasciola hepatica	Metacercari a	ingestio n	hermaphrodit e	Operculated egg	Triclabendaz ole

SCHISTOSOMIASIS HEMATOBIUM

Causes squamous cell carcinoma ,terminal hematuria


terminal spine

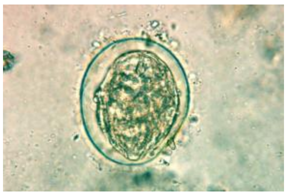
SCHISTOSOMIASIS MANSONI

Causes - inferior mesenteric plexus


lateral spine

SCHISTOSOMIASIS JAPONICUM

Causes - superior mesenteric plexus



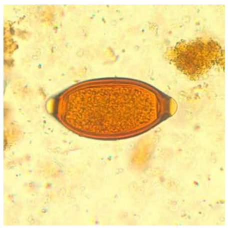
FMGE 2024,2026

Cestodes	**Definite**	**Intermediate host**
Taenia solium	man	Pig
Taenia saginata	man	Cattle
Echino Granulosus	dog	sheep /man
H. nana	man	Man

Diphyllobothrium latum (↓ B12 absorption)	man	1st- cyclops 2nd- Fish

NEMATODES

TRICHURIS TRICHURIA/WHIPWORM



-barrel/dumb bell shaped egg
-Cause Fe deficiency anaemia
-Rectal prolapse seen (coconut cake like rectum)

ENTEROBIUS VERMICULARIS

NEET PG 25,24

Perianal pruritus +
Not present in stool sample
National institute health swab - scotch tape method

SKIN PENETRATION SEEN IN **NEET PG 24, FMGE 2026**
Strongyloides (Rhabdiform larvae image)
Ancylsotoma
Schistosoma

EGGS FLOAT ON SATURATED SALT SOLUTION **FMGE 2021**
F- fertilised eggs of ascaris
A- ancylostoma duodenale
T- trichuris trichiura
E- enterobius vermicularis
H- H.nana

AUTOINFECTION SHOWN BY
C- cryptosporidium
H- H.nana
E- enterobius
S- strongyloides
T- T.solium

Balantidium coli **Doc : tetracycline**	

VIROLOGY

One liners-

Most of the DNA are ds virus except parvo
Most of the RNA are ss virus except reovirus
Most of the DNA virus replicate in nucleus except Pox virus
Most of the RNA virus replicate in cytoplasm except influenza
Largest size virus - pox virus
smallest size virus - parvo virus

Shapes of viruses Bullet shape- rabies Star shape- astro virus Wheel shape- rotavirus Brick shape- pox virus Space vehicle- adenovirus

DNA VIRUS

Herpes viridae
Hepadna virus,HBV
Adenovirus
Papilloma virus
Pox virus
SV40 virus
JC virus

Cowdry type A	**Cowdry type B**
Herpes virus Yellow fever	Adeno virus Polio virus

VARICELLA ZOSTER/CHICKEN POX VIRUS

SAR- >90%
infectivity period- 2 days before ←– rash → 5 days after (until the scab fails)
Centripetal rash present, Dew drop appearance
Incubation period- 14-15 days

HZ OTICUS/RAMSY HUNT SYNDROME

bell's palsy

vesicles over EAC,TM and cheeks → geniculate ganglion is involve

HZ ophthalmicus

single sided lesion present on face dermatological involvement (T3-L3)

The ophthalmic branch of the trigeminal nerve is involved.

CMV

NEET PG 23,24

Congenital CMV syndrome-

C3M2V1

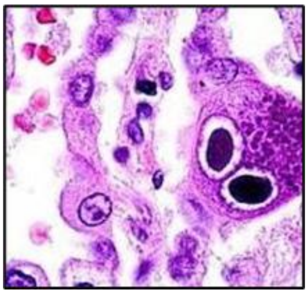
chorioretinitis calcifications convulsions

microcephaly mental retardation

periventricular calcifications

Differential diagnosis- toxoplasmosis

in toxoplasmosis parenchymal ventricular calcification is seen



CMV retinitis- DOC- Ganciclovir

EBV

Affect B cells → leads to polyclonal proliferation of B cells

EBV attach via CD21/CR2 receptor

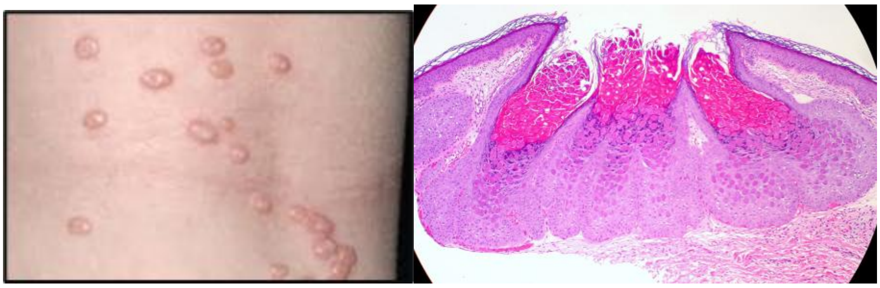
Most common malignancy caused by EBV → Gastric carcinoma

POX VIRUS

molluscum contagiosum

pearly umbilicated nodule over the skin/genitals present

on biopsy- henderson peterson bodies (intracytoplasmic bodies)



PARVO VIRUS

FMGE 2021

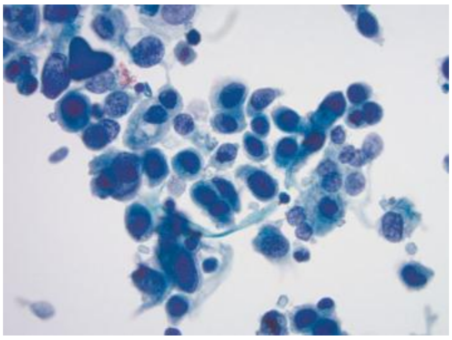
ssDNA virus
Non envelope
mc affect RBC - cause aplastic crisis in sickle cell anaemic and pure red cell aphasia
can be seen also associated with 5th day disease - Erythema infectiosum
Slapped cheek appearance present - gloves and socks syndrome

JC VIRUS

Cultured in human fetal glial cell
JC virus → crosses BBB → infects oligodendrocytes → causes demyelination → leads to PML (Progressive Multifocal Leukoencephalopathy)

BK VIRUS

Causes - post kidney transplant infection
DECOY CELLS are seen in BK nephropathy



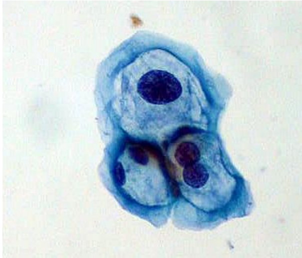
HUMAN PAPILLOMA VIRUS

FMGE 2021,25

L1 capsid → used in vaccine
E2- controller region
E6- inactivates products of P53 gene
E7- inactivates products of Rb gene

Cutaneous - low risk - 1,2,3 → filiform wart,verruca plantaris

- high risk - 5,8 → epidermodysplasia verruciformis



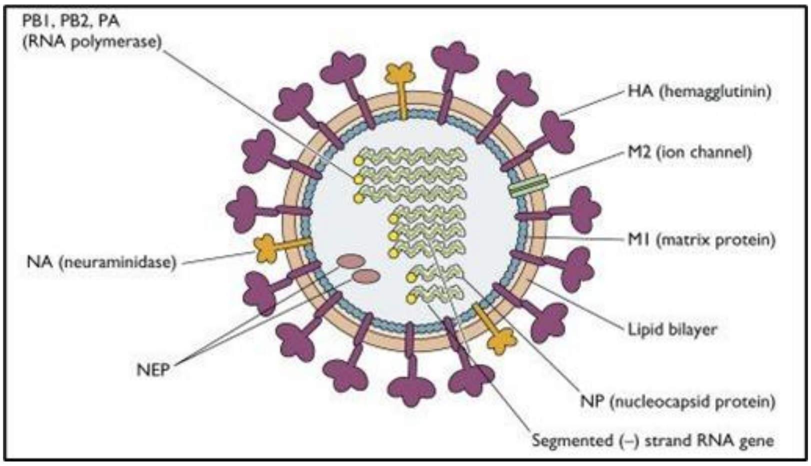
Mucosal - low risk - 6,11 → condyloma accuminata ,recurrent laryngeal papillomas
- high risk - 31,33,16,18 → squamous cell carcinoma

HPV most commonly causes oropharyngeal cancer
Diagnosis →Histopathology → E4/E5 raisanoid nucleus

INFLUENZA VIRUS A, B, C

FMGE 2022,25

glycoprotein H - hemagglutinin – adheres to cell
glycoprotein N - neuraminidase – receptor destroying enzyme
Antigenic shift-
abrupt, drastic and discontinuous process due to reassortments
Antigenic drift-
single point mutation and continuous in nature



COXSACKIE VIRUS A

Causes -
Flaccid paralysis
Acute hemorrhagic conjunctivitis
Hand food and mouth disease

MUMPS

FMGE 2021

mc presentation- B/L parotitis
IP- 2- 3 weeks
lab diagnosis- PCR/ELISA for mumps virus
Vaccine strain- Jeryl lynn strain

ENTEROVIRUS

Enterovirus 70- acute hemorrhagic conjunctivitis
Enterovirus 71- HFMD
Enterovirus 72- hepatitis A

MEASLES

NEET PG 21,fmge 2026

Enveloped SSRNA
most characteristic feature- koplik's spots (opposite to lower 2nd molar)
SAR - >90%
IP - 10-14 days
Inclusion bodies- warthin finkeldey giant cells

Clinical features-
cough, coryza, conjunctivitis
chronic complication - SSPE
Vaccine strain - Edmonston Zagreb

RUBELLA

IP- 14 days
Forchheimer spots seen on uvula blueberry muffin rash seen
Congenital rubella syndrome
triad-
Cataract PDA SNHL

RABIES VIRUS

FMGE 2025

Bullet shape
Decrease acetylcholine at various sites and include
Neural cell apoptosis
Dynein protein and Retrograde axonal transport
Negri bodies seen in hippocampus + cerebellum
Best method for antemortem diagnosis → Direct fluorescent antibody test

HEPATITIS

Property	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Virus structure	naked,RNA	Envelope, DNA	Envelope, RNA	Envelope, RNA	naked,RNA
Transmission	Fecal-oral	Parenteral, sexual	Parenteral, sexual	Parenteral, sexual	fecal-oral
Incubation period	15-50	50-150	15-150	50-150	15-50
Other diseases	None	HCC, cirrhosis	HCC, cirrhosis	Cirrhosis,fulminant hepatitis	none

MCC of fulminant hepatitis- Hepatitis D
MCC of fulminant hepatitis in pregnancy- hepatitis E

ROTA VIRUS

NEET PG 23

wheel with spoke appearance
dsRNA virus
MCC of diarrhoea in children
vaccine side effect- intussusception
NSP-4 → responsible for secretory diarrhea
Dx- antigen detection in stool sample

Apart from CD4 tell cell count in HIV patient + SOB + fever
if chest x ray finding is given mark according to x ray

X Ray finding-
Lobe consolidation - pneumonia

Snowstorm pattern- TB

Perihilar opacities- pneumocystis carinii